

CH 5: DIMENSION REDUCTION METHODS

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Dimensionality Reduction

- **Dimensionality Reduction** refers to the process of reducing the number of random variables (features) under consideration, by obtaining a set of principal or more informative variables.
- It can be done through:
 - **Feature Selection** – selecting a subset of original features.
 - **Feature Extraction** – transforming data into a lower-dimensional space (e.g., PCA).
- It helps in simplifying models, improving performance, and enabling visualization of high-dimensional data.

Curse of Dimensionality

- Coined by Richard Bellman, it refers to the **problems and challenges that arise when analyzing and organizing data in high-dimensional spaces**.
- As the number of features increases:
 - Volume of space increases **exponentially**.
 - Data points become sparse; **distance metrics lose meaning**.
 - Classifiers (e.g., k-NN) suffer due to **overfitting and computational complexity**.
- Example: In high dimensions, all points tend to become equidistant, making classification less reliable.

Need for Dimensionality Reduction

- **Improves Model Performance:** Reduces noise and irrelevant features, leading to better generalization.
- **Reduces Computation Time and Memory:** Fewer features mean faster processing and simpler models.
- **Helps in Visualization:** Allows plotting data in 2D or 3D after reducing from higher dimensions.
- **Avoids Overfitting:** By eliminating redundant or correlated features, models become less complex and more robust.
- **Enhances Interpretability:** Simplifies data to a form that's easier to understand and analyze.

Fisher Discriminant Analysis (FDA)

Fisher Discriminant Analysis is a supervised learning technique used for both dimensionality reduction and classification. The key idea behind FDA is to find a projection (a linear combination of features) that best separates two or more classes.

FDA transforms high-dimensional data into a lower-dimensional space, such that the projected points of different classes are as far apart as possible, while the points within the same class are as close together as possible.

This is especially helpful in pattern recognition problems like:

- Face recognition
- Handwritten digit recognition
- Medical diagnosis
- Document classification

Purpose of FDA in Classification

FDA is not just a dimensionality reduction method, it is built specifically to aid in classification tasks also.

- Helps to **separate different classes** in data clearly.
- Finds a **best line or direction** to project high-dimensional data.
- Makes **classification easier** after reducing dimensions.
- Uses **class labels** to guide the reduction (supervised method).
- Keeps **important features** that help tell the classes apart.
- Removes **unnecessary or noisy features**.
- Works well for both **dimensionality reduction** and **classification**.
- FDA helps in projecting data to a space where classification becomes easier and more accurate.

Supervised Dimensionality Reduction

FDA is a **supervised** technique, meaning it uses **label information** (class information) during the learning process. This contrasts with unsupervised methods like PCA, which ignore class labels.

Key points:

- **Supervised:** It uses both the input features and the class labels to determine how to best reduce dimensions.
- **Class Means:** FDA calculates the mean vectors of each class to understand their distributions.
- **Scatter Matrices:** It constructs scatter matrices:
 - Within-class scatter matrix (SW): Measures how samples spread within a class.
 - Between-class scatter matrix (SB): Measures how far apart the class means are.
- FDA then finds a projection that maximizes the ratio of between-class scatter to within-class scatter.

Steps of FDA

- ① Compute Class Means — for each class C_i

$$\mu_i = \frac{1}{N_i} \sum_{x \in C_i} x$$

- ② Compute the Within-Class Scatter matrix

for each class,
$$S_i = \sum_{x \in C_i} (x - \mu_i)(x - \mu_i)^T$$

Total within-class
Scatter matrix is
given by

$$S_W = \sum_i S_i$$

- ③ Compute Between-Class Scatter Matrix

for 2 classes,
$$S_B = (\mu_1 - \mu_2)(\mu_1 - \mu_2)^T$$

- ④ Solve Generalized Eigenvalue

$$S_W^{-1} S_B w = \lambda w$$

λ = eigen values
 w = eigen vectors
(Fisher projection vector)

- ⑤ Select top k eigenvalues — to form a projection matrix W , after sorting them in descending order

- ⑥ Project the Original Data — new data repr

$$y = W^T x$$

⑦ w can also be calculated like,
$$w = S_W^{-1} (\mu_1 - \mu_2)$$

Problem 1: Two-Class Fisher Linear Discriminant

Given the following two classes of 2D data points:

- Class 1:

$$x_1 = \begin{bmatrix} 2 \\ 3 \end{bmatrix}, x_2 = \begin{bmatrix} 3 \\ 4 \end{bmatrix}, x_3 = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$$

- Class 2:

$$x_4 = \begin{bmatrix} 5 \\ 1 \end{bmatrix}, x_5 = \begin{bmatrix} 6 \\ 2 \end{bmatrix}, x_6 = \begin{bmatrix} 7 \\ 3 \end{bmatrix}$$

Tasks:

- Compute the class means.
- Compute the within-class scatter matrix S_W .
- Derive the optimal projection vector w using Fisher's criterion.
- Project all points onto w and determine the threshold for classification.

Answer

(a) Class Means

$$\mu_1 (\text{for class 1}) = \frac{1}{3} \left(\begin{pmatrix} 2 \\ 3 \end{pmatrix} + \begin{pmatrix} 3 \\ 4 \end{pmatrix} + \begin{pmatrix} 4 \\ 5 \end{pmatrix} \right) = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$$

$$\mu_2 (\text{for class 2}) = \frac{1}{3} \left(\begin{pmatrix} 5 \\ 1 \end{pmatrix} + \begin{pmatrix} 6 \\ 2 \end{pmatrix} + \begin{pmatrix} 7 \\ 3 \end{pmatrix} \right) = \begin{pmatrix} 6 \\ 2 \end{pmatrix}$$

(b) Within-Class Scatter matrix

for S_1

$$(x_1 - \mu_1) = \begin{pmatrix} 2 \\ 3 \end{pmatrix} - \begin{pmatrix} 3 \\ 4 \end{pmatrix} = \begin{pmatrix} -1 \\ -1 \end{pmatrix}$$

$$(x_2 - \mu_1) = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad (x_3 - \mu_1) = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$S_1 = \begin{pmatrix} -1 \\ -1 \end{pmatrix} \begin{pmatrix} -1 & -1 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \end{pmatrix} \begin{pmatrix} 0 & 0 \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \end{pmatrix}$$
$$= \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix}$$

for S_2

$$(x_4 - \mu_2) = \begin{pmatrix} -1 \\ -1 \end{pmatrix} \quad (x_5 - \mu_2) = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad (x_6 - \mu_2) = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$S_2 = \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix}$$

$$(S_w)_{\text{Total within class scatter matrix}} = \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix} + \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix} = \begin{pmatrix} 4 & 4 \\ 4 & 4 \end{pmatrix}$$

(c) Btwn-Class Scatter matrix

$$\mu_1 - \mu_2 = \begin{pmatrix} 3 \\ 4 \end{pmatrix} - \begin{pmatrix} 6 \\ 2 \end{pmatrix} = \begin{pmatrix} -3 \\ 2 \end{pmatrix}$$

$$S_B = \begin{pmatrix} -3 \\ 2 \end{pmatrix} \begin{pmatrix} -3 & 2 \end{pmatrix} = \begin{pmatrix} 9 & -6 \\ -6 & 4 \end{pmatrix}$$

$$\boxed{S_w^{-1} S_B w = \lambda w} \quad \text{or} \quad \boxed{w = S_w^{-1} (\mu_1 - \mu_2)}$$

We see S_{w0} not invertible

$$\text{as } |S_{w0}| = 4.4 - 4.4$$

$\therefore$ we consider a fixed w value, $w = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

Class 1 Projections.

$$y_1 = \begin{bmatrix} 2 \\ 3 \end{bmatrix} \begin{bmatrix} 1 & -1 \end{bmatrix} = -1$$

$$y_2 = \begin{bmatrix} 3 \\ 4 \end{bmatrix} \begin{bmatrix} 1 & -1 \end{bmatrix} = -1$$

$$y_3 = \begin{bmatrix} 4 \\ 5 \end{bmatrix} \begin{bmatrix} 1 & -1 \end{bmatrix} = -1$$

Class 2 Projections.

$$y_4 = \begin{bmatrix} 5 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & -1 \end{bmatrix} = 4$$

$$y_5 = \begin{bmatrix} 6 \\ 2 \end{bmatrix} \begin{bmatrix} 1 & -1 \end{bmatrix} = 4$$

$$y_6 = \begin{bmatrix} 7 \\ 3 \end{bmatrix} \begin{bmatrix} 1 & -1 \end{bmatrix} = 4$$

① Threshold

$$\text{Mean proj of class 1} = \frac{-1 + (-1) + (-1)}{3} = -1$$

$$\text{Mean proj of class 2} = \frac{4 + 4 + 4}{3} = 4$$

$$\text{Mean threshold} = \frac{-1 + 4}{2} = 1.5$$

Conclusion

If a class has projections $y = w^T x < 1.5$ then Class 1
else Class 2.

Problem 2: Fisher LDA with Unequal Covariance Structures

Suppose you are given two classes of 3D data where the within-class scatter matrices

$$\begin{aligned} \bullet \quad S_{W1} &= \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ \bullet \quad S_{W2} &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix} \end{aligned}$$

The class means are:

$$\bullet \quad \mu_1 = \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}, \quad \mu_2 = \begin{bmatrix} 6 \\ 7 \\ 3 \end{bmatrix}$$

Tasks:

- Compute the total within-class scatter $S_W = S_{W1} + S_{W2}$.
- Find the Fisher discriminant vector $\mathbf{w}$.
- Determine the direction in which data should be projected for maximal separation.
- Discuss the implication of unequal scatter matrices on the linear separability.

Answer

Scatter Matrices:

$$S_{W1} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad S_{W2} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

Class Means:

$$\mu_1 = \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}, \quad \mu_2 = \begin{bmatrix} 6 \\ 7 \\ 3 \end{bmatrix}$$

- ✓ (a) Compute total within-class scatter S_W

$$S_W = S_{W1} + S_{W2} = \begin{bmatrix} 2+1 & 0 & 0 \\ 0 & 2+1 & 0 \\ 0 & 0 & 1+2 \end{bmatrix} = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

- ✓ (b) Compute Fisher Discriminant Vector $\mathbf{w}$

$$\mathbf{w} = S_W^{-1}(\mu_2 - \mu_1)$$

$$\mu_2 - \mu_1 = \begin{bmatrix} 6 \\ 7 \\ 3 \end{bmatrix} - \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \\ 2 \end{bmatrix}$$

Since S_W is diagonal:

$$S_W^{-1} = \begin{bmatrix} 1/3 & 0 & 0 \\ 0 & 1/3 & 0 \\ 0 & 0 & 1/3 \end{bmatrix}$$

$$\mathbf{w} = S_W^{-1}(\mu_2 - \mu_1) = \begin{bmatrix} 1/3 & 0 & 0 \\ 0 & 1/3 & 0 \\ 0 & 0 & 1/3 \end{bmatrix} \begin{bmatrix} 4 \\ 4 \\ 2 \end{bmatrix} = \begin{bmatrix} 4/3 \\ 4/3 \\ 2/3 \end{bmatrix}$$

- ✓ (c) Direction for Maximum Class Separation

The optimal projection direction is along the Fisher vector $\mathbf{w}$:
This is the axis onto which all 3D data should be projected
for best linear class separation.

$$\mathbf{w} = \begin{bmatrix} 4/3 \\ 4/3 \\ 2/3 \end{bmatrix}$$

- ✓ (d) Implication of Unequal Scatter Matrices

- Unequal scatter matrices (i.e., $S_{W1} \neq S_{W2}$) imply that covariance is class-dependent.
- Fisher LDA assumes common covariance, so when this is violated:
 - The linear boundary may not be optimal.
 - Classes might be better separated by a quadratic (nonlinear) boundary — handled by Quadratic Discriminant Analysis (QDA).
- Still, Fisher's method gives a reasonable linear approximation by averaging the scatter matrices.

Problem 3: Generalization to Multi-Class LDA

Given 3 classes with the following class means:

$$\bullet \mu_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \mu_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \mu_3 = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

Assume each class has a within-class scatter matrix:

$S_{W_i} = I$ (2x2 identity matrix) and there are equal number of samples in each class.

Tasks:

- Derive the between-class scatter matrix S_B .
- Compute the total within-class scatter matrix S_W .
- Solve the generalized eigenvalue problem for $S_W^{-1} S_B$.
- Determine the LDA projection matrix and dimensionality of reduced space.

Answer

$$\bullet \text{ Class means: } \mu_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \mu_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \mu_3 = \begin{bmatrix} -1 \\ 0 \end{bmatrix} \quad \bullet \text{ Within-class scatter for each: } S_{W_i} = I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

✓ a) Between-class scatter matrix S_B

Let μ be the overall mean:

$$\mu = \frac{1}{3}(\mu_1 + \mu_2 + \mu_3) = \frac{1}{3} \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} + \begin{bmatrix} -1 \\ 0 \end{bmatrix} \right) = \frac{1}{3} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ \frac{1}{3} \end{bmatrix}$$

Now compute $S_B = \sum n_i(\mu_i - \mu)(\mu_i - \mu)^T$

For μ_1 :

$$\mu_1 - \mu = \begin{bmatrix} 1 \\ 0 \end{bmatrix} - \begin{bmatrix} 0 \\ \frac{1}{3} \end{bmatrix} = \begin{bmatrix} 1 \\ -\frac{1}{3} \end{bmatrix} \Rightarrow (\mu_1 - \mu)(\mu_1 - \mu)^T = \begin{bmatrix} 1 & -\frac{1}{3} \\ -\frac{1}{3} & \frac{1}{9} \end{bmatrix}$$

For μ_2 :

$$\mu_2 - \mu = \begin{bmatrix} 0 \\ 1 \end{bmatrix} - \begin{bmatrix} 0 \\ \frac{1}{3} \end{bmatrix} = \begin{bmatrix} 0 \\ \frac{2}{3} \end{bmatrix} \Rightarrow (\mu_2 - \mu)(\mu_2 - \mu)^T = \begin{bmatrix} 0 & 0 \\ 0 & \frac{4}{9} \end{bmatrix}$$

For μ_3 :

$$\mu_3 - \mu = \begin{bmatrix} -1 \\ 0 \end{bmatrix} - \begin{bmatrix} 0 \\ \frac{1}{3} \end{bmatrix} = \begin{bmatrix} -1 \\ -\frac{1}{3} \end{bmatrix} \Rightarrow (\mu_3 - \mu)(\mu_3 - \mu)^T = \begin{bmatrix} 1 & \frac{1}{3} \\ \frac{1}{3} & \frac{1}{9} \end{bmatrix}$$

Add all:

$$S_B = \begin{bmatrix} 1 & -\frac{1}{3} \\ -\frac{1}{3} & \frac{1}{9} \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & \frac{4}{9} \end{bmatrix} + \begin{bmatrix} 1 & \frac{1}{3} \\ \frac{1}{3} & \frac{1}{9} \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & \frac{2}{3} \end{bmatrix}$$

✓ c) Generalized Eigenvalue Problem $S_W^{-1} S_B$

We need to solve for the eigenvalues λ and eigenvectors v of the matrix: $\begin{bmatrix} \frac{2}{3} & 0 \\ 0 & \frac{2}{9} \end{bmatrix} v = \lambda v$

This is a diagonal matrix, so the eigenvalues are simply the diagonal entries: $\lambda_1 = \frac{2}{3}, \lambda_2 = \frac{2}{9}$

And the corresponding eigenvectors are: $v_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, v_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

✓ d) Determine the LDA Projection Matrix and Dimensionality of Reduced Space

The LDA projection matrix is formed by the eigenvectors corresponding to the largest eigenvalues. Since the largest eigenvalue is $\lambda_1 = \frac{2}{3}$, we choose the corresponding eigenvector $v_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$.

Thus, the LDA projection matrix is:

$$\mathbf{W} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

The dimensionality of the reduced space is determined by the number of non-zero eigenvalues. Since we are working with a 2-dimensional space and we have 2 eigenvalues, the dimensionality of the reduced space is 1 (as we select the largest eigenvalue and its corresponding eigenvector).

Principal Component Analysis (PCA)

Principal Component Analysis (PCA) is a widely used statistical method for reducing the dimensionality of large datasets while retaining as much information (variance) as possible. It is an unsupervised technique that transforms the data into a new coordinate system where the axes (principal components) are ordered based on the variance they capture. PCA is especially useful in exploratory data analysis and for visualizing high-dimensional data. **Key Concepts in PCA:**

- **Principal Components (PCs):** The new axes in the transformed space. These components are the eigenvectors of the covariance matrix, and each successive component captures the remaining variance orthogonally.
- **Eigenvalues:** Measure the variance captured by each principal component. Larger eigenvalues correspond to principal components that capture more variance in the data.
- **Dimensionality Reduction:** PCA reduces the dimensionality of the data by selecting only the top k eigenvectors. This allows for a more compact representation of the data.
- **Variance Explained:** The ratio of the variance captured by a given principal component relative to the total variance. The first principal component captures the most variance, the second captures the second-most variance, and so on.

Applications of PCA:

- **Data Visualization:** PCA is often used to reduce the dimensionality of high-dimensional data for visualization. For example, reducing data to 2 or 3 dimensions allows us to plot complex data in a 2D or 3D scatter plot.
- **Noise Reduction:** By discarding components with small eigenvalues (which correspond to less variance), PCA can be used to remove noise from data, leading to better performance in machine learning models.
- **Feature Extraction:** PCA can extract features that summarize the data efficiently. The new features (principal components) are linear combinations of the original features and capture the most important patterns in the data.
- **Preprocessing for Machine Learning:** PCA is commonly used as a preprocessing step to reduce the dimensionality of data before feeding it into machine learning algorithms, particularly when dealing with high-dimensional data such as images or gene expression data.

Advantages of PCA:

- **Reduces Dimensionality:** PCA helps reduce the number of features in a dataset while retaining most of the information, which can improve the performance of machine learning models.
- **Improves Efficiency:** By reducing the number of features, PCA can make the computations more efficient and decrease the time required for training machine learning models.
- **Handles Correlation:** PCA can handle correlated features effectively by transforming the data into a new coordinate system where the features are uncorrelated.

Disadvantages of PCA:

- **Loss of Interpretability:** The principal components are linear combinations of the original features, so it can be difficult to interpret what the new features represent.
- **Sensitive to Scaling:** PCA is sensitive to the scale of the features, so it is essential to standardize the data before applying PCA if the features have different units or variances.
- **Linear Assumptions:** PCA assumes that the relationships between features are linear, which may not always be true for complex datasets.

EXAMPLE:

Find the Principal Component of each point (X1,X2) using PCA

F	Ex 1	Ex 2	Ex 3	Ex 4
X ₁	4	8	13	7
X ₂	11	4	5	14

Answer

Mean

$$\bar{X}_1 = \frac{4+8+13+7}{4} = 8 \quad \bar{X}_2 = \frac{11+4+5+14}{4} = 8.5$$

Covariance Matrix

$$\text{Covariance Matrix (Cov)} = \begin{bmatrix} \text{Cov}(X, X) & \text{Cov}(X, Y) \\ \text{Cov}(Y, X) & \text{Cov}(Y, Y) \end{bmatrix}$$

$$\begin{aligned} \text{Cov}(X, X) &= \frac{1}{N-1} \sum (X - \bar{X})(X - \bar{X}) = \frac{1}{N-1} \sum (X - \bar{X})^2 \\ &= \frac{1}{3} ((4-8)^2 + (8-8)^2 + (13-8)^2 + (7-8)^2) \\ &= \frac{1}{3} (16 + 0 + 25 + 1) = 14 \end{aligned}$$

$$\begin{aligned} \text{Cov}(X, Y) &= \frac{1}{N-1} \sum (X - \bar{X})(Y - \bar{Y}) \\ &= \frac{1}{3} ((4-8)(11-8.5) + (8-8)(4-8.5) + (13-8)(5-8.5) + (7-8)(14-8.5)) \\ &= \frac{1}{3} (-4 \times 2.5 + 0 + 5(-3.5) + (-1)(5.5)) \\ &= \frac{1}{3} (-10 - 17.5 - 5.5) = -11 \end{aligned}$$

$$\text{Cov}(Y, X) = -11$$

$$\begin{aligned} \text{Cov}(Y, Y) &= \frac{1}{N-1} \sum (Y - \bar{Y})^2 = \frac{1}{3} ((11-8.5)^2 + (4-8.5)^2 + (5-8.5)^2 + (14-8.5)^2) \\ &= \frac{1}{3} (6.25 + 12.25 + 12.25 + 30.25) = 23 \end{aligned}$$

$$\therefore \text{Cov} = \begin{bmatrix} 14 & -11 \\ -11 & 23 \end{bmatrix}$$

Eigen values

$$\begin{vmatrix} 14-\lambda & -11 \\ -11 & 23-\lambda \end{vmatrix} = 0$$

$$\Rightarrow 322 - 37\lambda + \lambda^2 - 121 = 0$$

$$\Rightarrow \lambda^2 - 37\lambda + 201 = 0$$

$$\lambda = \frac{37 \pm \sqrt{1369 - 804}}{2} = \frac{37 \pm 23.7697}{2}$$

$$= 30.3849, 6.6152$$

$$\text{Eigen value } \lambda_1 = 30.3849$$

$$\lambda_2 = 6.6152$$

Eigen vector

where,

$$\begin{bmatrix} 14-\lambda & -11 \\ -11 & 23-\lambda \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = 0 \quad U = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} \text{ is eigen vector}$$

we choose largest eigen value λ_1

$$\therefore \begin{bmatrix} 14-30.3849 & -11 \\ -11 & 23-30.3849 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = 0$$

$$\Rightarrow \begin{bmatrix} -16.3849 & -11 \\ -11 & 7.3849 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = 0$$

$$\Rightarrow \begin{bmatrix} -16.3849u_1 & -11u_2 \\ -11u_1 & 7.3849u_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\text{Let } -16.3849u_1 = 11u_2 = t$$

$$\Rightarrow \frac{u_1}{11} = \frac{u_2}{-16.3849} = t$$

$$\therefore U = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 11 \\ -16.3849 \end{bmatrix}$$

$$\text{To find unit eigen vector, } |U| = \sqrt{11^2 + 16.3849^2} = 19.7349$$

$$\therefore \text{Unit eigen vector} = \frac{1}{19.7349} \begin{bmatrix} 11 \\ 16.3849 \end{bmatrix} = \begin{bmatrix} 0.5574 \\ -0.8302 \end{bmatrix}$$

(e)

Principal Component evaluation

X	Y	(X - $\bar{x}$)	(Y - $\bar{y}$)
4	11	-4	2.5
8	4	0	-4.5
13	5	5	-3.5
7	14	-1	5.5

$\therefore$ To find principal component,

$$e^T (x - \bar{x})$$

$$\therefore y_1 = \begin{bmatrix} 0.5574 & -0.8302 \end{bmatrix} \begin{bmatrix} -4 \\ 2.5 \end{bmatrix} = -4.3051$$

$$y_2 = \begin{bmatrix} 0.5574 & -0.8302 \end{bmatrix} \begin{bmatrix} 0 \\ -4.5 \end{bmatrix} = 3.7359$$

$$y_3 = \begin{bmatrix} 0.5574 & -0.8302 \end{bmatrix} \begin{bmatrix} 5 \\ -3.5 \end{bmatrix} = 5.6927$$

$$y_4 = \begin{bmatrix} 0.5574 & -0.8302 \end{bmatrix} \begin{bmatrix} -1 \\ 5.5 \end{bmatrix} = -5.1235$$

Feature	PCA (Principal Component Analysis)	FDA / LDA (Fisher Discriminant Analysis)
Type	Unsupervised	Supervised
Uses Class Labels?	No	Yes
Goal	Maximize total variance	Maximize class separability
Based On	Covariance matrix	Between-class & within-class scatter matrices
Projection Direction	Directions of maximum variance	Directions that best separate classes
Application Focus	Feature extraction, compression, visualization	Classification, pattern recognition
Data Distribution Assumption	No distribution assumption	Often assumes Gaussian distribution per class
Principal Components	Orthogonal	Not necessarily orthogonal
Best For	Unlabeled data, reducing redundancy	Labeled data, enhancing discrimination

Principal Component Analysis (PCA) – What it Does:

1. Identifies directions of maximum variance in the dataset.
2. Projects data onto a new axis (principal components) without considering class labels.
3. Reduces dimensionality by keeping the components that capture the most variance.
4. Compresses data while preserving as much information as possible.
5. Helps in visualization, noise reduction, and feature extraction.
6. Works well when class separation is not the goal, just structure simplification.

Fisher Discriminant Analysis (FDA / LDA) – What it Does:

1. Finds projection axes that maximize class separation.
2. Uses both between-class scatter and within-class scatter to compute discriminant directions.
3. Projects data onto a lower-dimensional space where classes are more distinguishable.
4. Reduces dimensions to (number of classes – 1).
5. Primarily used for classification tasks.
6. Assumes labeled data and often assumes normal distribution within each class.